
MIRRORNET: CAN MEDICAL IMAGE ANONYMIZATION REALLY PROTECT PATIENT IDENTITY?

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ABSTRACT

Medical images are routinely de-identified—names, dates, and other metadata removed—and then shared for research, teaching, and public benchmarks under the assumption that this renders them anonymous. Such de-identification protects the metadata but not the pixels, and—apart from scans that directly contain facial structures—whether the image content itself identifies the patient has received little scrutiny. We investigate this question by learning a cycle-consistent correspondence between a cross-sectional medical image and a non-medical, patient-identifying image, using a pair of coupled, cycle-consistent variational autoencoders. From a held-out scan, the model recovers a recognisable likeness of the patient (identity-region MAE = 0.163); conversely, it synthesises a scan from such an image. These results indicate that a de-identified medical scan remains identifying—it is, in effect, a photograph of the patient—and that imaging data should be governed as biometric data rather than as anonymisable records. To support reproducibility, the code and trained models are shared at <https://github.com/attilasimko/public-repository>.

Keywords cross-modal estimation · domain transfer · clinical-image confidentiality · generative models · computed tomography

1 Introduction

Medical images are shared at scale—across institutions, in teaching collections, and in the open benchmarks that drive methodological progress—after *de-identification*: the removal of names, dates, and header fields that tie an image to a record. The governing assumption is that a de-identified image is anonymous, and it underpins the release of large public imaging datasets and the routine exchange of scans for secondary use. It is equally embedded in everyday hospital workflows: anonymised images are routinely shared for second opinions, referrals, multi-site studies, and archival, on the understanding that anonymisation has severed the link to the patient. But de-identification operates on metadata, whereas the diagnostic content lives in the pixels, and it is far from obvious that the pixels are anonymous. A 2D cross-sectional medical image captures detailed, person-specific anatomy; if that anatomy encodes identity in a recoverable form, then stripping the header does not anonymise the image, and every shared scan is, to some degree, a disclosure of the patient. This concern is not particular to one modality or one region of the body—it applies wherever an image carries identity-linked structure.

Whether the pixels are identifying is an empirical question, and modern generative models make it a pressing one. They reconstruct detailed, plausible images from weak conditioning, and translate freely between imaging modalities. The most direct expression of identity is a likeness—the canonical biometric and the form in which a person is most directly recognised. We therefore ask directly whether a de-identified scan can be turned back into a recognisable identifying image. A positive answer would show that anonymisation by metadata removal is insufficient: that a medical image carries identity in the signal itself, and that sharing it is closer to sharing a photograph of the patient than to releasing an anonymous record.

The presented work builds on generative image-to-image translation, or domain transfer, which is an active field in deep learning for medical imaging: variational autoencoders and adversarial networks routinely map one acquisition to another, for example synthesising computed tomography from magnetic resonance (MR). In the usual setting source

and target share a field of view and can be spatially registered, so skip connections carry local detail and the task reduces largely to appearance transfer. Our setting differs in acquisition geometry. A photograph is a perspective projection: light from a 3D scene converges through a single optical centre onto a plane. A cross-sectional medical image is not a projection but a slice—a 2D sample of a 3D volume, in which every pixel is a physical location in an axial plane through the body. The two therefore share no common frame of reference and cannot be brought into correspondence by standard image registration. Any information that crosses between the domains must instead pass through a low-dimensional latent code, which makes the model an information bottleneck by construction.

If the content of a de-identified scan is identifying, the consequences reach beyond any single image. Medical images are archived and shared in large volumes on the assumption that de-identification renders them anonymous; under that reading, each such release is a potential disclosure of the individuals depicted. The ability to extract patient information from a scan would place cross-sectional imaging alongside fingerprints and photographs as identifying biometric data, with consequences for how consent is obtained, how public datasets are released, and for what de-identification can be assumed to guarantee.

2 Materials and Methods

Our primary goal is to recover a patient-identifying image from a cross-sectional medical scan—to reconstruct, from a CT slice, a recognisable image of the person. We learn this mapping ($B \rightarrow A$) jointly with its inverse ($A \rightarrow B$, which synthesises a scan from an identifying image) as a single cycle-consistent system, so that the two are constrained to be mutually invertible. In our experiments the identifying image (domain A) is a facial photograph of the patient and the medical image (domain B) is an axial pelvic CT slice.

2.1 Data

The medical images are a subset of the pelvic CT volumes from the SynthRAD2023 Grand Challenge dataset [1], comprising 180 patients acquired at two centres. Volumes are stored in LPS orientation at $1 \times 1 \times 2.5$ mm. Because CT intensities are physical and standardised, we apply a single fixed Hounsfield-unit (HU) window of $[-1000, 1500]$ across the whole cohort and rescale it to $[0, 1]$ (no per-volume normalisation). For each volume we retain the central axial band (the central 40% of body-bearing slices), transpose each slice to the conventional radiological axial view, and pad—rather than crop—each slice to a square before resampling to 128×128 , preserving the full lateral field of view. We also carry the provided body mask through the identical geometry, for use at evaluation. This yields 8580 axial slices.

The identifying images are facial photographs drawn from the UTKFace dataset [2], restricted to the adult male subset via the provided age and gender annotations. Each photograph is letterboxed (aspect-ratio preserving) to 128×128 RGB.

Each scan is *enrolled* with an identifying image; the association is fixed once, is bijective, and is frozen for all experiments. Because the imaging subjects are themselves de-identified and no photograph of them exists, the enrolled image is a *surrogate* identity: a stand-in that lets us demonstrate the mechanism without exposing any real person. Because the learned mapping is invertible and cycle-consistent, the recovered image is the canonical partner of a given scan rather than an arbitrary sample. Patients are partitioned into training, validation, and test sets (70/15/15) at the *patient* level and stratified by acquisition centre, so that no slice of a patient appears in more than one partition, preventing slice-level leakage between neighbouring axial positions of the same subject.

2.2 Architecture

Each direction is a convolutional variational autoencoder. A four-stage strided encoder maps the RGB input $x \in \mathbb{R}^{3 \times 128 \times 128}$ to the parameters of a diagonal Gaussian posterior $q_\phi(z | x) = \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x))$ over a d -dimensional latent code ($d = 256$), and a symmetric four-stage decoder maps a sample z to the estimated output $\hat{y} \in \mathbb{R}^{1 \times 128 \times 128}$ through a sigmoid. We deliberately use no skip connections between encoder and decoder: the latent code is the sole information channel between the domains, so that outputs reflect the learned prior rather than pixel-level copying of a spatially unrelated input.

Let y be the target. Each direction minimises a reconstruction and a latent-regularisation term,

$$\mathcal{L} = \underbrace{\|\hat{y} - y\|_1}_{\text{reconstruction}} + \beta \underbrace{\text{KL}(q_\phi(z | x) \| \mathcal{N}(0, I))}_{\text{latent regularisation}}, \quad (1)$$

with a deliberately small $\beta = 0.01$ (a large KL weight collapses the posterior to the prior, so that every input decodes to the same mean output; the small weight forces the latent to carry the input). The two directions are then coupled by a cycle-consistency term, described next.

We couple two models of the architecture above: $F_{B \rightarrow A}$, mapping a CT to an identifying image (the primary task), and $F_{A \rightarrow B}$, mapping an identifying image to a CT, each with the appropriate input and output channels. $F_{B \rightarrow A}$ is supervised on the enrolled image, and we additionally add a cycle-consistency term (weight $\lambda_{\text{cyc}} = 1$), $\|F_{B \rightarrow A}(F_{A \rightarrow B}(a)) - a\|_1$ for $a \in A$ and $\|F_{A \rightarrow B}(F_{B \rightarrow A}(b)) - b\|_1$ for $b \in B$, so that an identifying image round-tripped through a CT (and a CT through an identifying image) is preserved. This invertibility is what lets us speak of *the* identifying image associated with a scan: the mapping is one-to-one by construction, so the recovered identity is determined by the scan rather than drawn at random from a prior.

2.3 Training

Both directions are trained jointly with AdamW (lr = 2×10^{-4}), batch size 32, on a single NVIDIA RTX 2080 Ti (11 GB). Horizontal flips and mild intensity jitter are applied to the target during training; this augmentation is one of the ablated components.

2.4 Evaluation and ablation

The primary task, $B \rightarrow A$ recovery of the identifying image, is evaluated by the *identity-region MAE*—the mean absolute error between the reconstructed and the enrolled image—and a *perceptual distance*, the L_1 distance between their VGG16 features; both are computed on a central crop so that the letterbox background, which carries no identity, does not contribute. For the secondary $A \rightarrow B$ synthesis we follow common practice for synthetic CT and report the mean absolute error (MAE) in Hounsfield units *inside the body mask*, both overall and broken down by tissue class (air < −200 HU, soft tissue −200 to 200 HU, bone > 200 HU). Each metric is the mean over the test set, and in each results column the best value is set in *italic*.

We characterise MirrorNet with an ablation that tests the significance of three factors, each scored by the quality of the $A \rightarrow B$ (CT) output. The first is the stochastic control R —a non-negative integer that fixes the stochastic-initialisation regime of a run and selects among nominally equivalent configurations rather than any capacity or loss setting—swept over $R \in \{0, 1, 2\}$. The second is the cycle-consistency coupling between the two directions, and the third is the target augmentation; each is removed in turn from the strongest- R configuration, which is adopted as MirrorNet.

3 Results

We report the two directions in turn—first the recovery of an identifying image from a scan, then the synthesis of a scan from an identifying image—and close with the cycle experiments that relate them.

The model $F_{B \rightarrow A}$ maps a held-out CT slice directly to an identifying image, and the reconstruction attains an identity-region MAE of 0.163 and a perceptual distance of 14.0. Figure 1 shows, for two held-out subjects, the input CT, the enrolled (true) identifying image, and the identifying image recovered from the scan alone. The main characteristics of each subject—gender, hair, build, apparent age, and, notably, skin tone—are well recovered from the pelvic scan, reproducing precisely the attributes that a stripped-header, cross-sectional image is assumed not to carry. Notably, facial hair, and facial expressions are not recovered well: this is consistent with the two data points having been acquired at different timepoints, between which such transient features may plausibly differ, whereas temporally stable biometric attributes persist and are recovered. We return to this limitation below.

The reconstruction is stable across the test set rather than the product of a few favourable cases: the reported metrics are means over all held-out subjects, and the qualitative behaviour in Figure 1 is representative. Because the two subjects were never seen in training and are paired to their scans only through the surrogate enrolment, the recovered attributes must be inferred from the scan itself rather than memorised. A reconstruction of this quality does not need to be photorealistic to be damaging: a recognisable likeness, even an approximate one, is enough to narrow a scan to a small set of candidate identities or to confirm a suspected match, which is all that a re-identification attack requires.

As a secondary result, the same coupled model runs in the $A \rightarrow B$ direction, synthesising a CT from an identifying image. Table 1 reports the ablation (Section 2.4)—the R sweep and the removal of each MirrorNet component—giving body-masked synthesis quality in HU, overall and per tissue class. The sweep shows that increasing R to $R=2$ improves the estimate. Removing cycle-consistency degrades it, and removing augmentation degrades it further. Figure 2 shows $A \rightarrow B$ estimates for two held-out subjects, alongside a Grad-CAM saliency map indicating which regions of the input drive the estimate. Based on the Grad-CAM results, the cheeks and the overall head shape appear to be the

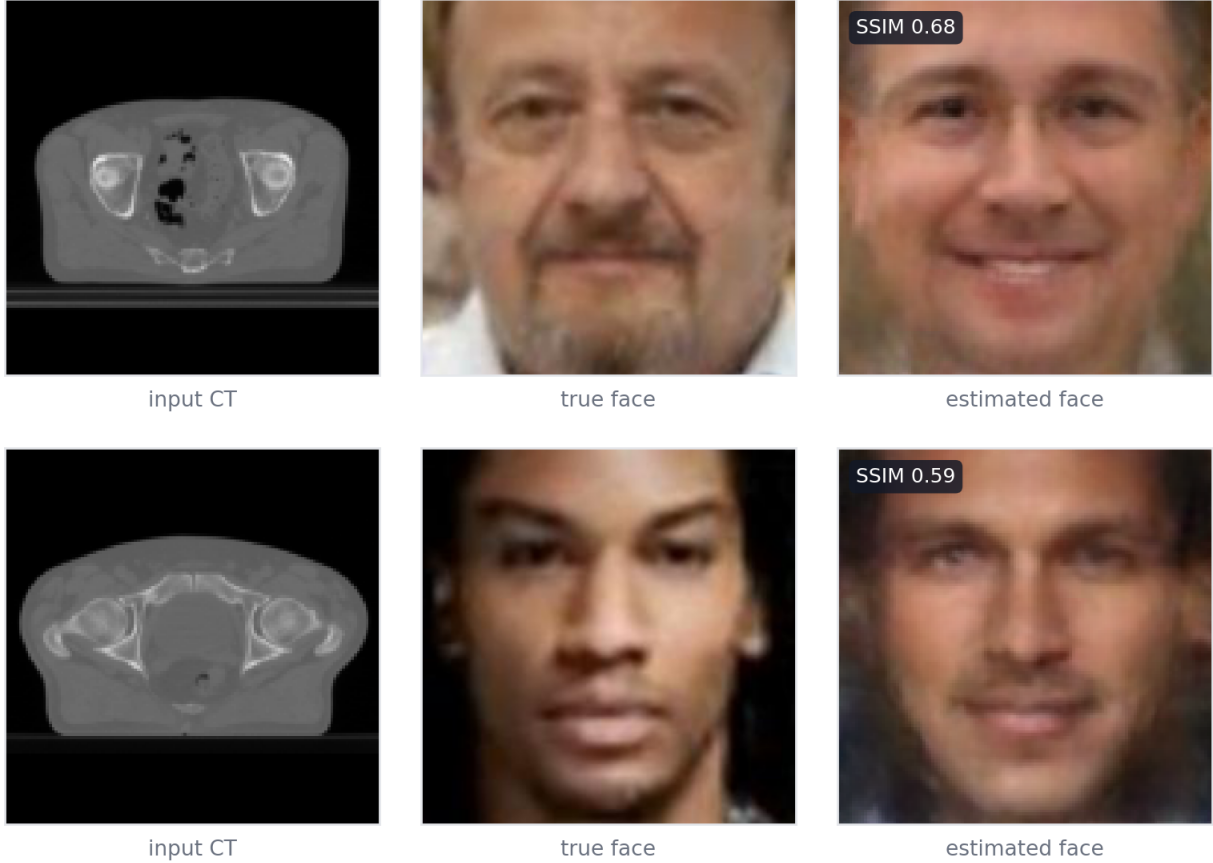


Figure 1: From a scan to an identity, on two held-out subjects. Each row: input CT, the enrolled (true) identifying image, and the identifying image recovered directly from the scan.

Table 1: Ablation over MirrorNet on the held-out test set (means only; best per column in *italic*). The upper rows sweep the stochastic control R ; the strongest is MirrorNet. The lower rows each remove one component of MirrorNet. MAE is in Hounsfield units, evaluated inside the body mask, overall and per tissue class.

Model	MAE _{body}	MAE _{air}	MAE _{soft}	MAE _{bone}
MirrorNet ($R=0$)	<i>181.9</i>	301.8	<i>134.0</i>	322.3
MirrorNet ($R=1$)	186.2	<i>279.5</i>	146.4	331.6
MirrorNet ($R=2$)	183.3	289.6	138.5	340.9
MirrorNet – cycle-consistency	186.7	303.6	138.9	339.8
MirrorNet – augmentation	183.3	301.2	136.5	<i>317.3</i>

most significant regions when reconstructing the CT slices. The current $A \rightarrow B$ estimates do not yet carry enough high-resolution anatomical detail to be used directly for diagnostic purposes, and improving their fidelity to that standard remains future work.

Finally, because $F_{A \rightarrow B}$ and $F_{B \rightarrow A}$ are trained jointly under a cycle-consistency constraint, each domain can be round-tripped through the other. A photograph taken to a CT and back to an identifying image is preserved to an identity-region MAE of 0.148, and a CT taken to an identifying image and back to a CT to a body-masked MAE of 108 HU (Figure 3). Both round-trips close with little loss in either crossing. This near-invertibility indicates that, under the learned correspondence, the two representations encode largely equivalent information.

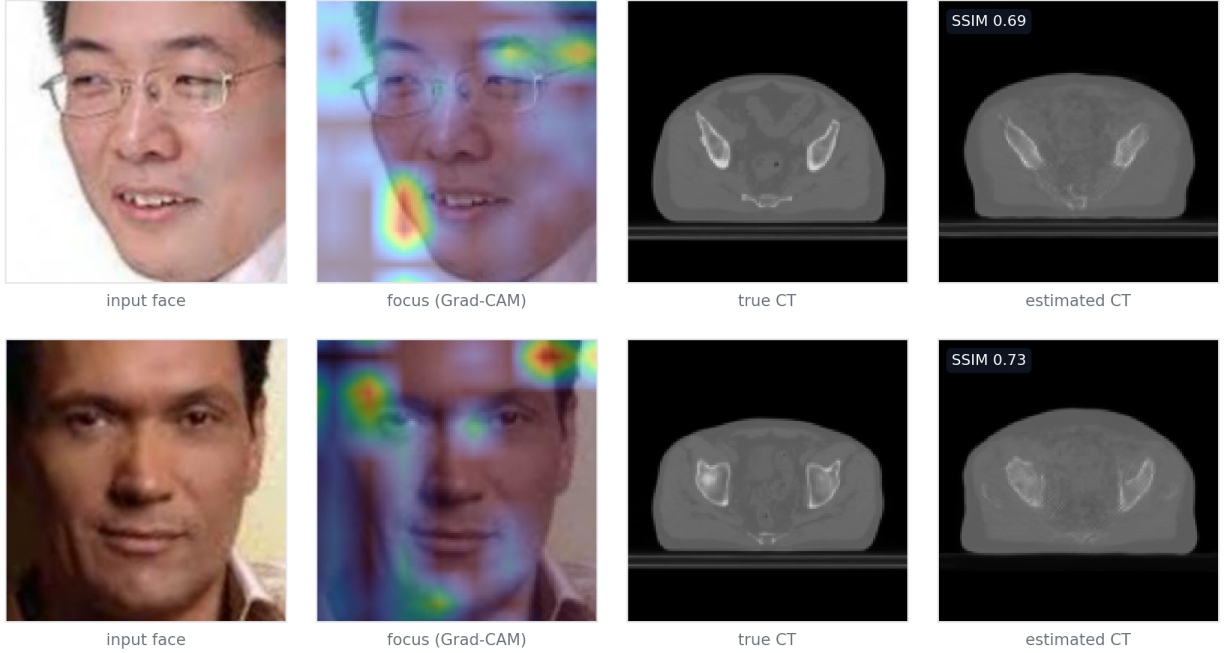


Figure 2: Forward estimates from MirrorNet on two held-out subjects. Each row: input photograph, a Grad-CAM saliency map over the photograph, the ground-truth CT, and the MirrorNet estimate.

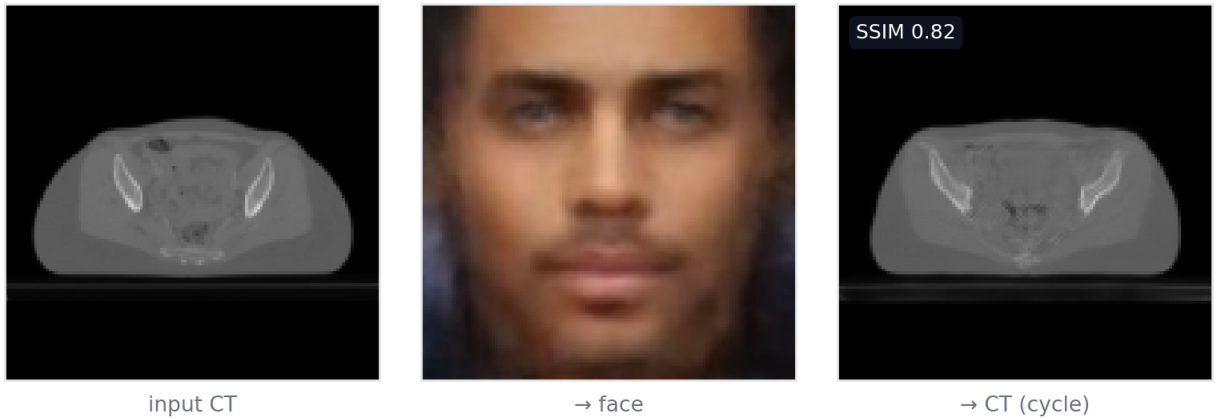


Figure 3: A full round-trip on a held-out subject: input CT, the intermediate recovered identifying image, and the CT after the complete cycle (CT→image→CT). The near-recovery of the domain after the round-trip underlies the interchangeability claim.

4 Discussion

MirrorNet establishes an invertible, cycle-consistent correspondence between an ordinary photograph and a cross-sectional medical image: a scan yields back an identifying image, and an identifying image yields a plausible, anatomically coherent scan. The two directions have different implications, considered in turn.

Because the correspondence is invertible, a CT is not an anonymous volume of Hounsfield units but a representation from which an identifying image can be recovered. De-identification that strips names, dates, and accession numbers leaves the pixel data untouched, and it is the pixel data that is identifying. On this evidence the boundary between “medical” and “biometric” data, on which much of health-data governance rests, does not hold for cross-sectional imaging: such an image is biometric. Consent obtained for imaging is not consent to release an identifying image; an

anonymised public imaging archive may therefore constitute a release of the depicted individuals, and the risk increases as further anatomical regions, modalities, and longitudinal scans of the same person accumulate.

The $A \rightarrow B$ direction admits a contrasting reading: if a photograph already determines a plausible scan, downstream steps such as positioning, planning, and triage could in principle be initialised from a photograph. This reading is in tension with the preceding one: a scan cannot be at once information-poor enough to be replaced by a photograph and information-rich enough to reconstruct an identifying image. The clinical implications of the $A \rightarrow B$ direction are beyond the scope of this study.

The study is restricted to 2D axial slices at 128×128 resolution and to a single anatomical region. The ablation identified the stochastic control R as an important hyperparameter to tune. The recovery is also not uniform across attributes: stable characteristics of a subject are reproduced, but transient ones—facial expression in particular—are not, and reconstructions tend toward a neutral expression. Several directions follow from this. The most immediate is to move from single slices to full 3D volumes, and to recover an identifying image from a complete scan rather than one axial plane; using the whole volume should sharpen the recovered identity and, in the reverse direction, yield a synthetic anatomy that is coherent in three dimensions rather than slice by slice. A second direction is to extend beyond the pelvis to other anatomies—the head, chest, and abdomen—to test whether the effect holds wherever a scan carries person-specific structure, and to measure how much identity each region leaks. Further work includes additional imaging modalities, longitudinal linkage across a patient’s repeated scans, and a formal threat model that quantifies the re-identification risk.

5 Ethics and data availability

Open medical datasets such as SynthRAD2023 are a prerequisite for this study and for reproducible research more broadly. The present results nonetheless bear on their release: if a scan can be inverted to an identifying image, de-identified imaging is not anonymous, and the sharing of such data cannot rest on de-identification alone. The argument extends to the conditioning modality, in that a public photograph can be mapped to a plausible cross-sectional anatomy, so the release of ordinary identifying images is also a potential route to sensitive inference. These considerations apply in principle to every dataset used in this work, including the artefacts released with it. All identifying images used here are drawn from a public, consented corpus and all scans are de-identified, so no real identity is exposed. The code and trained models are available at <https://github.com/attilasimko/public-repository>.

6 Conclusions

We presented MirrorNet, an invertible, cycle-consistent system that estimates an axial cross-sectional medical image from an ordinary photograph and, conversely, recovers an identifying image from a scan. The $A \rightarrow B$ direction relates to photograph-initialised imaging workflows; the $B \rightarrow A$ direction indicates that a de-identified scan is nonetheless identifying, and that cross-sectional imaging should be governed as biometric data. Enrolling scans with consented surrogate identities keeps the demonstration free of real-identity exposure. The full pipeline is released to support further study of cross-modal estimation, domain transfer, and their confidentiality implications.

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